Grasp Recovery Progress Summary

Project: Agent-aware supervision for a frozen RoboTwin VLA **Snapshot:** 12 November 2026

Objective

The work validates a VLA-first recovery path for failed mug grasps by either arm. The frozen Neotix policy remains responsible for normal contact-rich execution. Deterministic skills intervene only after a verified miss, place the robot in a safe retry state, and then return control to a fresh VLA invocation. No policy retraining or arbitrary 14-DoF planner output is introduced.

Current recovery contract

> Verified failed grasp → discard stale VLA actions → ground `mug_recovery_clearance` from the current mug pose → execute collision-aware `MOVE_TO` → verify endpoint → require `release_ready=true` and `mug_toppled=false` → open the cleared gripper → reset VLA history → fresh monitored VLA retry.

The same state machine is used for left-arm acquisition and right-arm pickup. No arm locking or wrist-rotation heuristic is used. A toppled mug or confirmed workspace loss terminates the episode instead of attempting an unsafe retry.

Implemented work

Area	Implementation
Failure monitoring	Event-based failed-grasp monitoring distinguishes inherited analytic contact from a new VLA closure attempt, avoiding premature retry failure.
Chunk handling	Remaining policy actions are discarded at intervention; every retry starts with reset policy history and fresh observations.
Grounded clearance	A symbolic pose is recomputed above the current mug from RGB-D/development evidence rather than using a hard-coded world pose.
Native motion	Recovery uses RoboTwin/CuRobo planning, measured TCP state, embodiment transforms, bounded waypoints, and native qpos execution.
Retry normalization	Clearance must complete within measured position/orientation tolerances before a support-gated gripper opening and VLA handback.
Evidence separation	Public perception remains planner-facing; simulator privilege is explicitly development-only and fills experimental evidence gaps.
Reproducibility	State-checked prefix replay, structured skill traces, 30 FPS two-camera video, and bounded T4 Modal configurations support repeatable experiments.

Evidence so far

- The vertical-clearance replay completed two recovery moves in 28 physics ticks each, with endpoint errors of 0.33 mm and 0.30 mm and orientation errors below 0.001 rad.
- The later grounded-waypoint replay completed both retry moves with endpoint errors of 0.02 mm and 0.65 mm. This confirmed the corrected TCP/end-link transform and the multi-waypoint path.
- The event monitor gave the VLA genuine post-recovery attempts instead of immediately classifying inherited contact loss as a new failed grasp.
- In the normalized live replay, the initial miss stopped after 77 policy actions and discarded three stale actions. Clearance finished in 78 physics ticks with 0.30 mm position error; the gripper then opened from 0.0 to 1.0. Fresh VLA inference confirmed a right-arm lift after 68 further actions.
- The complete CPU suite currently passes: 696 tests covering planner sequencing, skill contracts, guarded opening, perception, and left/right retry behavior.

Remaining gaps

1. Repeat the successful sequence across left and right failures and multiple seeds; report retry success, abort causes, endpoint error, and policy actions to recovery.
2. Replace remaining development-oracle evidence incrementally as public RGB-D predicates demonstrate equivalent reliability, especially support, toppling, workspace presence, and arm-specific lift.
3. Replay the verified contract through the Opus supervisor, ensuring its prompts select recovery only after an evidenced miss and never bypass the gates.
4. Validate hanging recovery separately; it is not established by grasp-retry success.

Current conclusion

The full right-arm recovery contract has now succeeded once in a recorded live replay: intervention, clearance, guarded opening, fresh inference, and verified lift. This establishes feasibility, not reliability. The next milestone is a multi-seed evaluation of both arms using public perception wherever it is reliable.